INT421—Digital Signal Processing

Nyquist–Shannon Sampling Theorem

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Topics of Last Lecture

- Signals
  - Speech, audio, light, radio, TV, radar, etc.
- Signal processing
  - Enhancement, restoration, reconstruction, synthesis, estimation, etc.
- Signal representation
  - Continues, discrete, and digital
- Basic DSP components
  - A/D converter, signal processor, D/A converter

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1. Signal Definition
2. Analog to Digital Conversion
3. System Definition
4. Summary

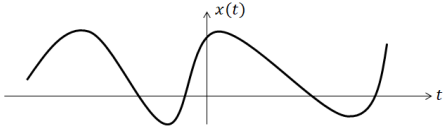
Signal Definition

- A function of one or more independent variables which conveys information on the nature of a physical phenomenon.
- Examples:
  - Temperature vs. time
  - Closing value of a stock market vs. day

Modeling

Physical signal is modeling by Mathematical function

- $x(t)$: signal of one independent variable t (e.g. time), x is called dependent variable.



Types of Signals

- **Real-valued** versus **complex-valued signals**
  - $s(t) = \cos(5t)$ is a real-valued signal. whereas
  - $s(t) = e^{j5t} = \cos(5t) + j\sin(5t)$ is a complex-valued signal
- **Single dimensional** versus **multi-dimensional signals**
  - Single dimensional signal $s(t) = \cos(5t)$ that are function of time variable t.
  - Multi-dimensional signals

$$S(t) = \begin{bmatrix} s_1(t) \\ s_2(t) \\ s_3(t) \end{bmatrix}$$

which is a vector or matrix of signal dimensional signals.

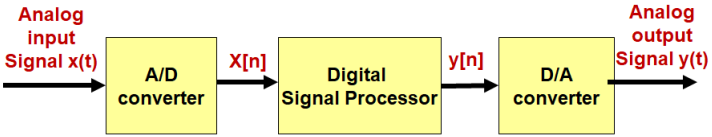
- Examples: antenna array beamformer signal.

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Basic Components of a DSP System

- DSP system consists of three main components, as illustrated below:



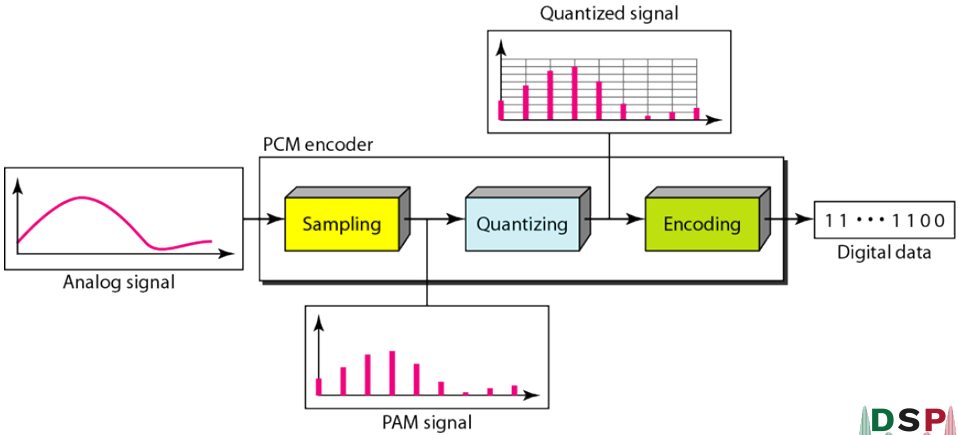
- The A/D converter transforms the analog signal $x(t)$ at the system input into a digital signal $x[n]$.
- The digital system performs the desired operations on the digital signal $x[n]$ and produces a corresponding output $y[n]$ also in digital form.
- The D/A converter transforms the digital output $y[n]$ into an analog signal $y(t)$ suitable for interfacing with the outside world.

Analog to Digital Conversion

- Discrete time signal often obtained by sampling a continuous-time signal. The analog-to-digital converter (ADC) consists of **four steps** to digitize an analog signal.
  - Filtering
  - Sampling
  - Quantization
  - Binary encoding
- Before we sample, we have to **filter** the signal to **limit the maximum frequency** of the signal as it affects the sampling rate.
- Filtering should ensure that we do not distort the signal, i.e. remove high frequency components that affect the signal shape.

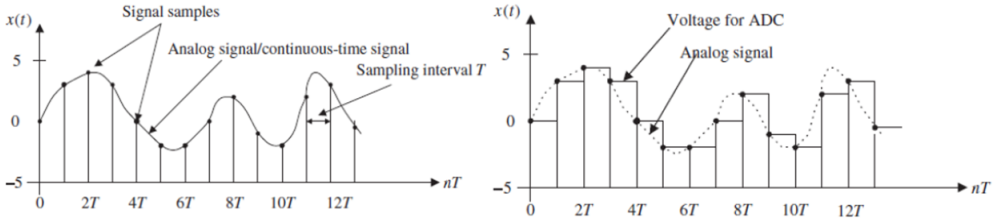
Analog to Digital Conversion

- A/D conversion can be viewed as a **three step process**



Analog to Digital Conversion

- **Sampling:**
  - Conversion from continuous-time, continuous valued signal to discrete-time, continuous-valued signal.
- **Quantization:**
  - Conversion from discrete-time, contentious-valued signal to discrete-time, discrete-time valued signal.
- **Coding:**
  - Conversion from discrete-time, discrete-valued signal to an efficient digital data format.



Nyquist–Shannon Sampling Theorem

Theorem

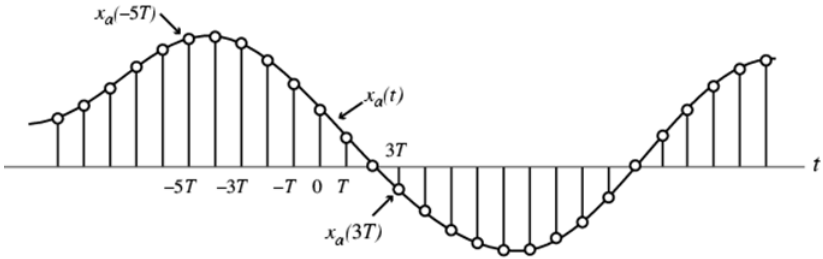
The sampling theorem guarantees that an analogue signal can be perfectly recovered as long as the sampling rate is at least twice as large as the highest-frequency component of the analogue signal to be sampled.

$$f_s \geq 2f_{max}$$

for a given sampling interval T, which is defined as the time span between two samples points, the sampling rate is therefore given by:

$$f_s = \frac{1}{T}$$

Nyquist–Shannon Sampling Theorem



- Sequence $\{x[n]\} = x_a(nT), \quad n = \cdots -1, 0, 1, 2 \ldots$
- T=Sampling Period; $1/T$= Sampling Frequency.
- $\{x[n]\} = \{\ldots, -0.2, \underset{\uparrow}{2.2}, 1.1, -3.7, 2.9, \ldots\}$
- Arrow indicates where $n = 0$

Quiz-1

Example

if the sampling period is $T = 125\mu$ seconds. What is the *sampling rate* of the signal?

Quiz-1

Example

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Solution

The sampling rate $f_s = 1/125\mu s = 8000$ *samples per seconds (Hz).*

Quiz-2

Example

Find the sampling frequency of the following signal

$$s(t) = 3\cos(50\pi t) + 10\sin(300\pi t) - \cos(100\pi t)$$

Quiz-2

Example

Find the sampling frequency of the following signal

$$s(t) = 3\cos(50\pi t) + 10\sin(300\pi t) - \cos(100\pi t)$$

Solution

The signal components have frequencies, $f_1 = 25Hz$, $f_2 = 150Hz$ *and* $f_1 = 50Hz$, *Then the sampling rate will be*

$$f_s \geq 2 \times 150 = 300$$

samples per seconds (Hz).

Quiz-3

Example

For the analog signal $s(t) = 3\cos(70\pi t)$, find the Nyquist sampling rate, also determine the digital signal frequency and the digital signal

Quiz-3

Example

For the analog signal $s(t) = 3\cos(70\pi t)$, find the Nyquist sampling rate, also determine the digital signal frequency and the digital signal

Solution

- *The maximum frequency component in* $s(t)$ *is* $f_{max} = \frac{70\pi}{2\pi} = 35Hz$.
- *According to Sampling Theorem, we need a sampling rate of* $f_s = 2f_{max} = 70Hz$
- *The digital signal would have a frequency* $\omega = 2\pi \times \frac{35}{70} = \pi$
- *the digital signal can be represented by* $s[n] = 3\cos(\pi n)$

Quiz-4

Example

Consider the signal $x_a(t) = 10\cos(2\pi 100t) + 5\cos(2\pi 5000t)$ is to be sampled. Determine the Nyquist rate for this signal. If the signal is sampled at 4 Khz, will the signal be recovered from its samples?

Quiz-4

Example

Consider the signal $x_a(t) = 10\cos(2\pi 100t) + 5\cos(2\pi 5000t)$ is to be sampled. Determine the Nyquist rate for this signal. If the signal is sampled at 4 Khz, will the signal be recovered from its samples?

Solution

- *Clearly, this signal contain two cosine waves with* $f_1 = 100$ *and* $f_2 = 5000$. *Since the highest frequency in the given is* $f_{max} = 5000$ *Hz, the nyquist rate is given as:*

$$Nyqusit - rate = 2F_{max} = 2 \times 5000Hz = 10KHz$$

- *This signal is sampled at* $f_s = 4000Hz$ *when the signal* $x_a(t)$ *is sampled, we get* $x[n]$ *as,*

$$x_a[n] = 10\cos(2\pi\frac{f_1}{f_s}n) + 5\cos(2\pi\frac{f_2}{f_s}n) = 10\cos(2\pi\frac{1000}{4000}n) + 5\cos(2\pi\frac{5000}{4000}n) = 10\cos(2\pi\frac{1}{4}n) + 5\cos(2\pi\frac{5}{4}n)$$

The second term can be rearranged in above equation as $x_a[n] = 10\cos(2\pi\frac{1}{4}n) + 5\cos(2\pi n + 2\pi\frac{1}{4}n)$.
Hence we can write above equation as, $x_a[n] = 10\cos(2\pi\frac{1}{4}n) + 5\cos(2\pi\frac{1}{4}n) = 15\cos(2\pi\frac{1}{4}n)$

Quiz-5

Example

Determine whether the following discrete time signals are periodic or not? If periodic, determine the fundamental period.

1. $x[n] = \cos(0.01\pi n)$

Quiz-5

Example

Determine whether the following discrete time signals are periodic or not? If periodic, determine the fundamental period.

1. $x[n] = \cos(0.01\pi n)$

Solution

The discrete time signal is periodic if its frequency f_0 can be expressed as ratio of two integers (i.e. f_0 is rational)

$$f_0 = \frac{k}{N}$$

Here k is some integer and N is fundamental period. Here, $\omega = 0.01\pi$. Since $\omega = 2\pi f_0$, we have

$$f_0 = \frac{\omega}{2\pi} = \frac{0.01\pi}{2\pi} = \frac{1}{200}$$

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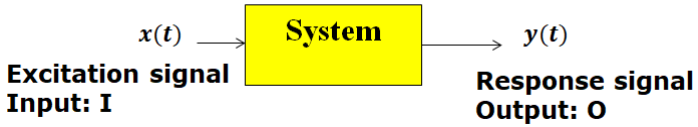
3 System Definition

4 Summary

System Definition

- An entity that manipulates one ore more signals to accomplish a **function** (job), thereby yielding **new signals**.
- Any **process or interaction of operations** that **transforms** an input signal into an output signal with properties different from those of the input.
- Examples:
  - Educational system
  - Transmitter

SISO: Single-Input Single-Output



Representation of Systems



x[n]: input, stimulus
y[n]: output, response

$$y[n] = S[\, x[n] \,]$$



Types of Systems

- Systems are classified corresponding to the number of inputs and outputs to:

MIMO: Multi-Input Multi-Output



MISO: Multi-Input Single-Output



SIMO: Single-Input Multi-Output



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Summary

- Fundamental signals
  - Unit sample, unit step, exponential, sinusoid
- System representation
  - A functional module between an input signal and an output signal
- Nyquist–Shannon Sampling Theorem

Any Questions?

